

A Practical Guide to Adversarial examples

TOPIC -- ADVERSARIAL EXAMPLES

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Secure learning algorithms Byzantine-resilient algorithms Multiple classifier systems AI-written algorithms. AIs that explore the training environment; for example, in image recognition, actively navigating a 3D environment rather than passively scanning a fixed set of 2D images. Privacy-preserving learning Ladder algorithm for Kaggle-style competitions Game theoretic models Sanitizing training data Adversarial training Backdoor detection algorithms Gradient masking/obfuscation techniques: to prevent the adversary exploiting the gradient in white-box attacks. This family of defenses is deemed unreliable as these models are still vulnerable to black-box attacks or can be circumvented in other ways. Ensembles of models have been proposed in literature, which have shown to be ineffective against evasion attacks but effective against data poisoning attacks. Moving target defense

-- 02 --

History At the MIT Spam Conference in January 2004, John Graham-Cumming showed that a machine-learning spam filter could be used to defeat another machine-learning spam filter by automatically learning which words to add to a spam email to get the email classified as not spam. In 2004, Nilesch Dalvi and others noted that linear classifiers used in spam filters could be defeated by simple "evasion attacks" as spammers inserted "good words" into their spam emails. (Around 2007, some spammers added random noise to fuzz words within "image spam" in order to defeat OCR-based filters.) In 2006, Marco Barreno and others published "Can Machine Learning Be Secure?", outlining a broad taxonomy of attacks. As late as 2013 many researchers continued to hope that non-linear classifiers (such as support vector machines and neural networks) might be robust to adversaries, until Battista Biggio and others demonstrated the first gradient-based attacks on such machine-learning models (2012–2013). In 2012, deep neural networks began to dominate computer vision problems; starting in 2014, Christian Szegedy and others demonstrated that deep neural networks could be fooled by adversaries, again using a gradient-based attack to craft adversarial perturbations. Further work would show that adversarial attacks are harder to produce in uncontrolled environments, due to the different environmental constraints that cancel out the effect of noise. For example, any small rotation or slight illumination on an

adversarial image can destroy the adversariality. In addition, researchers such as Google Brain's Nick Frosst point out that it is much easier to make self-driving cars miss stop signs by physically removing the sign itself, rather than creating adversarial examples. Frosst also believes that the adversarial machine learning community incorrectly assumes models trained on a certain data distribution will also perform well on a completely different data distribution. He suggests that a new approach to machine learning should be explored, and is currently working on a unique neural network that has characteristics more similar to human perception than state-of-the-art approaches. While adversarial machine learning continues to be heavily rooted in academia, large tech companies such as Google, Microsoft, and IBM have begun curating documentation and open source code bases to allow others to concretely assess the robustness of machine learning models and minimize the risk of adversarial attacks.

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Performance evaluation The choice of the evaluation metrics can impact the validity of the results. Having an inappropriate baseline involves failing to compare a new model against simpler, well-established baselines. Inappropriate performance measures means using metrics that do not align with the practical goals of the system. Reporting only "accuracy" is often described as insufficient for an intrusion detection system, where false-positive rates are considered critically important. Base rate fallacy is a failure to correctly interpret performance in the context of large class imbalances.

-- 04 --

Adversarial deep reinforcement learning Adversarial deep reinforcement learning is an active area of research in reinforcement learning focusing on vulnerabilities of learned policies. In this research area, some studies initially showed that reinforcement learning policies are susceptible to imperceptible adversarial manipulations. While some methods have been proposed to overcome these susceptibilities, in the most recent studies it has been shown that these proposed solutions are far from providing an accurate representation of current vulnerabilities of deep reinforcement learning policies.

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Feature robustness Researchers differentiate between features that can be easily manipulated and those that are more resistant to modification. For example, simple static attributes, such as header fields, may be altered by attackers, while structural features, such as control-flow graphs, are generally more stable but computationally expensive to extract.

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Concept drift Because malware creators continuously adapt their techniques, the statistical properties of malicious samples also change. This form of concept drift has been widely documented and may reduce model performance unless systems are updated regularly or incorporate mechanisms for incremental learning.

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Repeat this for $v_2, v_3, \dots$ until the desired level of error in the classifier is reached. It was discovered when the authors designed a simple baseline to compare with a previous black-box adversarial attack algorithm based on gaussian processes, and were surprised that the baseline worked even better.

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Data collection issues Security datasets vary across formats, including binaries, network traces, and log files. Studies have reported that the process of converting these sources into features can introduce bias or inconsistencies. In addition, time-based leakage can occur when related malware samples are not properly separated across training and testing splits, which may lead to overly optimistic results.

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System design and learning Feature engineering and training can cause issues. Data snooping is a common pitfall where a model is trained using information that would not be available in a real-world scenario. Spurious correlations result when a model learns to associate artifacts with a label, rather than the underlying security-relevant pattern. For example, a malware classifier might learn to identify a specific compiler artifact instead of malicious behavior itself. Biased parameter selection is a form of data snooping where model hyperparameters are tuned using the test set.

-- 10 --

Data poisoning Poisoning consists of contaminating the training dataset with data designed to increase errors in the output. Given that learning algorithms are shaped by their training datasets, poisoning can effectively reprogram algorithms with potentially malicious intent. Concerns have been raised especially for user-generated training data, e.g. for content recommendation or natural language models. The ubiquity of fake accounts offers many opportunities for poisoning. Facebook reportedly removes around 7 billion fake accounts per

year. Poisoning has been reported as the leading concern for industrial applications. On social medias, disinformation campaigns attempt to bias recommendation and moderation algorithms, to push certain content over others. A particular case of data poisoning is the backdoor attack, which aims to teach a specific behavior for inputs with a given trigger, e.g. a small defect on images, sounds, videos or texts.